

Deepam Sipani

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EDUCATION

B-Tech in Computer Science & Engineering – Data Science (GPA – 9.43)

2025-29

Dwarkadas J. Sanghvi College of Engineering

PROJECTS

Jewel Palace | [GitHub](#) | [Website](#)

Feb 2026 – Ongoing

- Engineered a full-stack e-commerce platform for an artificial jewellery business using the MERN stack (Next.js, Node.js, MongoDB), featuring secure Firebase authentication and a high-performance product catalog with image zoom.
- Integrated the Razorpay API to automate secure, end-to-end transactions, implementing server-side order generation and cryptographic signature verification to ensure robust and tamper-proof payment processing.
- Integrated the WhatsApp API to establish a real-time customer support channel, enabling direct communication between buyers and the shop owner for instant product inquiries.
- Overcome database complexity by designing a flexible MongoDB schema to manage varied jewellery attributes, ensuring data persistence and real-time inventory updates across user sessions.
- Optimized UX and reliability by implementing efficient state management and handling backend "cold starts," maintaining a professional interface that significantly boosted the business's online accessibility.

Itihasik | [GitHub](#) | [Website](#)

Nov 2025 – Jan 2026

- Architected a responsive historical documentation portal using a modular HTML5/CSS3 framework designed to host and organize large-scale cultural datasets.
- Developed a custom navigation system using semantic hierarchy to allow users to intuitively explore disparate historical periods and categorized information.
- Overcame layout instability by implementing a custom CSS Grid system that successfully handled varying content lengths and high-resolution media without breaking the UI.
- Refactored codebase for performance, shifting from inline styling to an externalized, dry (Don't Repeat Yourself) CSS architecture to improve maintainability and load speed.
- Eliminated cross-browser rendering bugs by standardizing flexbox properties and using vendor-specific prefixes, ensuring a uniform experience across all major web browsers.

Virtual Whiteboard | [GitHub](#)

Nov 2025 – Dec 2025

- Engineered a gesture-controlled virtual whiteboard using Python, OpenCV, and MediaPipe, enabling users to draw in 3D space via real-time hand tracking and webcam input.
- Developed a custom gesture-recognition engine to distinguish between "Pinch to Draw," "Fist to Erase," and "Open Hand for Idle" modes by calculating Euclidean distances between finger landmarks.
- Optimized real-time image processing by utilizing NumPy to manage a separate drawing canvas layer, overlaying it onto the video feed using weighted blending for a seamless AR experience.
- Solved jitter and tracking instability by implementing logic to handle landmark movement thresholds, ensuring smooth lines and preventing accidental strokes during rapid hand movements.
- Implemented high-accuracy spatial mapping by translating normalized hand coordinates into pixel-perfect canvas coordinates, maintaining drawing precision across different screen resolutions.

TECHNICAL SKILLS

Languages | JavaScript (ES6+), Python, HTML5, CSS3, C, Java, TypeScript

Frameworks | React, Node.js, Express.js, Next.js

Databases & Cloud | MongoDB, Firebase, Vercel, Render

Libraries | Pandas, NumPy, Matplotlib, TensorFlow, OpenCV, MediaPipe